

ELE Group 13

Joint Arm Mobile Locomotor JAM-L

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1 COMPONENT LIST & PURCHASED ITEMS

Components are listed in Table 1.

2 OUTSOURCED DESIGN/LIBRARY/SOFTWARE

No outsourced materials were used.

3 MECHANICAL DESIGN AND FABRICATION

JAM-L is a differential-drive mobile manipulator with a two tier 3D printed chassis, two skinny wheels, two front casters, and a three actuator serial arm ending with a custom geared parallel-jaw gripper.

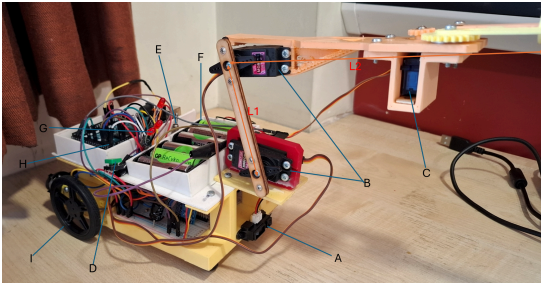


Fig. 1. J-MAL Robot . **A:** IR Distance Sensor, **B:** MG996 R servo, **C:** SG90 Servo, **D:** Push Button, **E:** RS Pro Toggle Switch, **F:** Battery Holder, **G:** LEDs, **H:** Arduino Mega, **I:** Skinny wheels attached to motor.

The most unique element is the geared gripper: the SG90 drives a 12-tooth pinion meshing with one of two 24-tooth spur gears (module 1.125, 20° pressure angle, 3 mm thick, 27 mm pitch diameter); the two large gears also mesh with each other, thus driving one counter rotates both, opening and closing the gripper fingers symmetrically. The 2:1 reduction doubles clamping torque at the cost of speed which is appropriate given the nature of the task (holding the marker). The arm uses two custom-length links ($L_1 = 81$ mm shoulder-to-elbow, $L_2 = 282$ mm elbow-to-tip) designed to reach the whiteboard from the IR-sensor stopping distance of 26.5 cm. The two-tier chassis carries batteries and the Arduino on the upper plate, motors and casters on the lower, with two front casters widening the support polygon thereby preventing lateral rocking of the robot.

The arm has 3 actuated DOF: shoulder pitch (MG996R), elbow pitch (MG996R), and gripper open/close (SG90 MicroServo). Only the two pitch joints position the end-effector in a vertical plane

(y–z); the mobile base adds 2 DOF (translation + yaw), giving 5 system DOF total (4 positioning + 1 gripper). The workspace in the arm's plane is an annulus with an outer radius 363 mm (fully extended), inner radius 201 mm (fully folded), centred on the shoulder pivot 105 mm above the floor. The demonstration drawing trajectory ($x = 235$ mm, $z = 90$ –230 mm) sits inside this region.

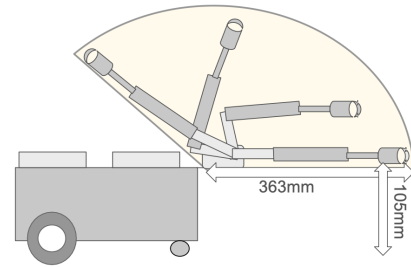


Fig. 2. Workspace of the JAM-L robot arm.

The elbow MG996R (11 kgfcm = 1.08 Nm stall torque at 6V) is the limiting actuator. With the forearm horizontal (worst case), the arm's self-weight (approximately 100 g, COM at $L_2/2$) creates 0.14 Nm, leaving 0.94 Nm for payload equivalent to 340 g at full extension. The marker weighs 18 g (5% of capacity), ensuring steady operation well below stall.

The support polygon is the quadrilateral formed by the two rear driven wheels (135 mm apart) and two front casters (150 mm forward of the drive axle). The total robot mass from Table 1 is approximately 968g. The chassis and electronics account for around 638 g with its centre of mass (COM) near the geometric centre of the chassis (75 mm forward of the drive axle), while the arm assembly accounts for approximately 330g. In the worst case (arm fully extended forward and horizontal) the arm COM sits approximately 222 mm forward of the shoulder. Taking moments about the drive axle, the system COM = $(0.638 * 75 + 0.330 * 222) / 0.968 = 125$ mm forward of the drive axle, which is 25 mm inside the front edge of the support polygon defined by the casters. Therefore, the robot remains statically stable at all arm configurations.

All custom parts were produced by **additive manufacturing (FDM)**. Additive was chosen

because nearly every part required out-of-plane features (servo cases, gear hubs, gripper jaws) that laser cutting cannot produce in one piece, and it avoids the material waste of machining. Total print time was around 48 hours with 10 parts. The total assembly took 4 hours. **PLA** was selected over ABS/PETG for its plant-based origin, industrial compostability, absence of toxic fumes, and lower printing temperatures which reduces energy usage. CAD validation in Fusion was used to minimise reprints.

The CAD modelled parts were printed using the FDM printers available at iForge at default settings: 0.20 mm layer height, 0.4 mm nozzle. The printer build volume (250*210*210mm) is the upper bound for the arm's reach [1].

4 ELECTRONICS

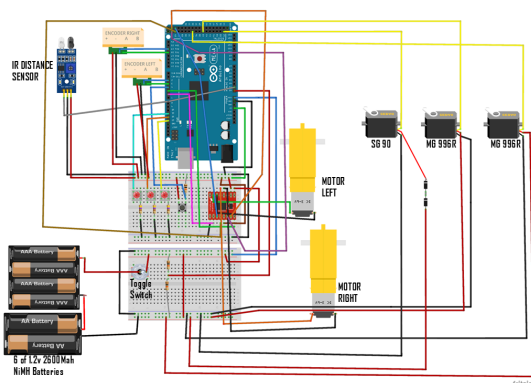


Fig. 3 Schematic of the electronic circuit.

- Microcontroller** - The Arduino Mega 2560 was selected over the Arduino Uno due to its expanded interrupt and pin capabilities. Full quadrature decoding of both wheel encoders requires four external interrupt pins; the Uno only provides two (pins 2 and 3), which would have limited encoder resolution. The Mega provides six external interrupt pins (pins 2, 3, 18, 19, 20, 21), accommodating all four encoder channels. The total I/O requirement — three servo PWM outputs, six motor driver pins, three LED indicator pins, one button, two analogue inputs, exceeded the Uno's capacity. The Mega operates at 5V logic and is powered via the VIN pin directly from the 7.2V NiMH battery, compatible with its onboard voltage regulator input range of 7–12V. A downside of the Mega is its larger physical size and higher idle current draw of approximately 70mA compared to smaller alternatives, however this was acceptable given the project requirements.

- Sensors** - The Sharp GP2Y0A21YK0F infrared distance sensor was used to detect proximity to the whiteboard and trigger the servo drawing sequence at a consistent repeatable distance. The smooth white surface of the whiteboard is an ideal IR reflector ensuring reliable readings. Key specifications: operating voltage 4.5–5.5V, current consumption 30mA, measurement range 10–80cm, analogue voltage output, update period 38ms. The sensor was operated in the 20–100cm range and connected to analogue pin A1.
- Actuators** - Two DFRobot FIT0450 brushed DC geared motors drive the wheels. With a 120:1 gearbox, 160RPM no-load speed at 6V, stall torque of 0.8kgf·cm, and no-load current of 0.17A, they provide sufficient torque to carry the arm assembly. The integrated quadrature encoders yield 3,840 counts per revolution in full quadrature, providing adequate resolution for the PI-controlled motion system. Motors are driven via the TB6612FNG dual H-bridge driver. Two MG996R servos actuate the base and elbow joints, selected for their high stall torque of 11kgf·cm at 6V. An SG90 micro servo actuates the gripper, its low weight of 9g and stall torque of 1.8kgf·cm is sufficient for light gripper actuation while minimising load on the elbow joint. Two 1N4007 rectifier diodes in series on the SG90 supply line provide a combined voltage drop of approximately 1.4V to prevent overdriving the micro servo.
- Circuitboard, Wires and connectors** - A breadboard was used for all connections. It hosts the voltage divider circuit for battery monitoring, LED indicator resistors, and power distribution rails for the 5V and 7.2V supply lines. The voltage divider uses two 10kΩ resistors to scale the 7.2V battery voltage down to a safe 3.6V for the Arduino's analogue input (A0), allowing real-time battery monitoring. Three LEDs with current-limiting resistors serve as a battery level indicator, illuminating progressively fewer LEDs as voltage drops below defined thresholds (7.2V, 7.0V, and 6.8V), providing a visual battery indication during operation. Dupont jumper wires were used throughout for all signal and power connections.
- Batteries and Operation Time** - JAM-L is powered by six 1.2V 2600mAh NiMH cells in series, giving a nominal voltage of 7.2V and total capacity of 18.72Wh. NiMH was selected for its stable discharge curve, rechargeability, and safety. The 7.2V nominal voltage is compatible with all subsystems and the weight of the pack is acceptable for the robot's stability. The system

operates in two distinct phases with non-overlapping high current loads: during navigation the drive motors draw approximately 0.50A combined while servos remain idle, giving a total of approximately 0.67A. During the drawing sequence the motors are stopped and the servos are active, drawing approximately 2.08A combined (1.80A for both MG996R servos under load and 0.25A for the SG90), giving a total of approximately 2.25A. Remaining loads like the Sharp GP2Y0A21YK0F IR sensor at 0.03A, Arduino Mega 2560 at 0.07A, battery indicator LEDs at 0.03A, and voltage divider at 0.001A — are present throughout both phases. At an average estimated load of approximately 1.5A across the full sequence the 2600mAh battery provides an estimated runtime of approximately 1.7 hours. A known limitation of NiMH is voltage sag under high current draw, managed by ensuring a full charge before each run.

5 CONTROL

JAM-L's motion control is implemented as a state machine, with each state representing a discrete phase of the robot's journey forward movements, turns, settle delays, servo actions, and the final drawing sequence. Transitions between states are triggered either by the robot reaching a target condition or a timer expiring.

Forward motion is governed by a PI controller that takes the error between a reference position (in cm) and the current position, calculated from the average of both wheel encoder counts converted to distance using the wheel circumference and encoder resolution of 3,840 counts per revolution. The control output is a PWM value applied to both motors. A separate straight-line correction term runs in parallel, computing the difference between left and right encoder counts and applying a proportional-integral correction to the individual motor PWM values to counteract drift and maintain a straight trajectory.

Turning is controlled by a PI controller operating on heading angle, derived from the differential of left and right encoder counts multiplied by the ratio of the wheel circumference to wheelbase giving us an angle in radians which is later converted to degrees. The reference angle is set for each turn in degrees and the controller drives the motors in

opposite directions until the angular error falls within a 2° tolerance.

For the final approach to the whiteboard, encoder-based position control is replaced by the Sharp GP2Y0A21YK0F IR distance sensor. The robot drives forward at a fixed PWM with straight-line correction active until the sensor reading falls below a threshold distance, at which point the motors stop and the servo drawing sequence is triggered. This ensures consistent stopping distance.

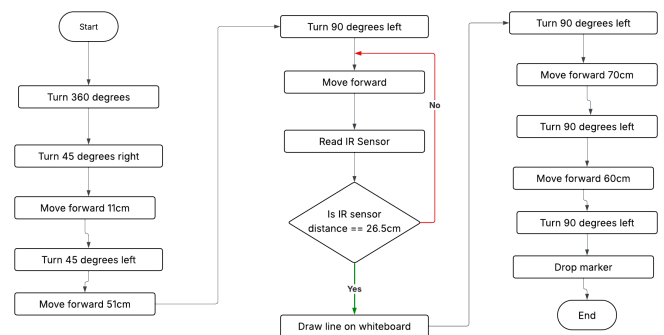


Fig. 4. Flowchart of the operational logic / main algorithm for Task 3.

The servo drawing sequence uses inverse kinematics to calculate joint angles for the base and elbow servos. Given a target Cartesian coordinate (x, z), the joint angles θ_2 and θ_3 are derived as:

$$\theta_2 = \text{atan2}(z, x) + \text{acos}((L_1^2 + r^2 - L_2^2) / (2 \cdot L_1 \cdot r)) - \pi/2$$

$$\theta_3 = \pi - \text{acos}((L_1^2 + L_2^2 - r^2) / (2 \cdot L_1 \cdot L_2)) - \pi/2$$

$$r = \sqrt{x^2 + z^2}$$

where : $L_1 = 81$, $L_2 = 282$ (mm)

An offset of $\pi/2$ is subtracted from each joint angle to reconcile the mathematical reference frame with the physical servo mounting orientation.

The total time for the entire sequence takes 1m33.

The custom designed geared parallel jaw gripper, which uses a 2:1 spur gear reduction to convert the SG90 MicroServo into a symmetric clamping mechanism, represents an advanced mechanical feature not standard in the kit. Moreover, the battery monitoring system with three tier LED indication, the IR-sensor-based final approach replacing encoder dead-reckoning, and the PI correction running in parallel with forward motion control show integration of multiple advanced control and sensing techniques.

Table 1: Mechanical and electronic components and material list

Component Name/Model	Count	Weight	Current/Power Consumption	Total Price excl. VAT	Link	Labels in Fig.1
MG 996R Servo	2	110g	1.8A	(included in the kit)	link	B
SG90 Servo	1	9g	0.25A	(included in the kit)	link	C
FIT0450 Motor	2	100g	0.50A(at 35% PWM)	(included in the kit)	link	I
TB6612FNG Motor Driver	1	negligible	negligible	(included in the kit)	link	not shown
Sharp GP2Y0A21YK0F IR Sensor	1	3.5g	0.03A	(included in the kit)	link	A
Arduino MEGA 2560 Rev 3	1	37g	0.07A	£0.00	link	H
NiMH 1.2V 2600mAh cells	6	180g	Power Source	£12.00	link	not labelled
3-Cell Battery Holder	2	30g	N/A	£3.00	link	F
BreadBoard	2	76g	N/A	(included in the kit)	link	-
1N4007 diodes	2	negligible	negligible	(provided from lab)	link	not shown
LEDs	3	negligible	0.06A	(included in the kit)	link	G
Resistors	6	negligible	negligible	(included in the kit)	link	not shown
Momentary Switch	1	negligible	negligible	(included in the kit)	link	D
RS Pro Toggle Switch	1	negligible	negligible	£2.30	link	E
Whiteboard Marker	1	18g	N/A	£0.10	link	not shown
Pololu 953 Ball Caster ½ inch	2	18g	N/A	£3.60	link	not labelled
Skinny Wheel	2	23.4g	N/A	(included in the kit)	link	I
Total 3D print Material PLA	-	362.95g	N/A	£10.89	link	-
	Total	967.85g	2.71A (excl. batteries)	£31.89		-

Table 2: Outsourced design/library/software materials

Material Name	Description	Link

(Pre-existing designs utilised in Fritzing or other PCB CAD software for Fig.2 need not be listed in Table 2, as it is not expected for students to create them from scratch.)

References

[1] iForge Makerspace. (2022). *3D Printers*. [online] Available at: <https://iforgesheffield.org/3d-printers/>.

Reflection on Group Project

As a group Our goal was to make the robot complete its task as simply as possible, which led us to initially rely solely on encoder feedback for navigation, as it appeared to be a straightforward solution. However, we underestimated the complexity of achieving reliable autonomous navigation in practice, as although encoders provide precise measurements in theory, their performance proved unreliable due to real-world factors such as wheel slip, variations in surface texture, and battery voltage sag. These issues introduced cumulative errors over time, particularly across longer distances, significantly reducing positional accuracy; as a result, we adapted our approach by integrating the infrared distance sensor for the final approach to the whiteboard, allowing the robot to correct accumulated error and achieve more consistent stopping behaviour, ultimately demonstrating the importance of combining multiple sensing methods for more robust navigation.

Rotimi Dayo We ran into a number of software issues so the use of the serial monitor was critical during debugging, as real-time feedback of encoder counts, position, angle, and voltage made it possible to identify and isolate faults that would have otherwise been difficult to detect. This experience reinforced the value of

systematic debugging, as systems rarely function as expected, and through continuous testing and refinement I developed a stronger understanding whilst also enhancing my coding, electronics, and debugging skills.

Valour Inyang

Throughout this project, I developed a stronger understanding of the processes involved in system design. I gained insight into the complexities of 3D printing and laser cutting, and I was particularly involved in designing the robot base using CAD, as well as working on the robot arm's inverse kinematics. I also expanded my knowledge of encoders, servos, and state machines. I actively participated in brainstorming sessions to explore ideas for the robot's functionality and contributed to debugging efforts aimed at resolving issues in the code. The project also helped me improve my interpersonal and communication skills.

Rustamkhon Ozodkhonzoda

I was in charge of the robot's mechanical design, making all of the unique components in Fusion, such as the servo mounts, arm links, chassis, and geared parallel-jaw gripper. My knowledge of CAD modelling has improved a lot as a result of this project, especially when it comes to creating pieces that need to correctly interact with commercially available parts such as servo horns. I learned how choosing correct gear geometry has a direct impact on practical performance from the gripper mechanism. It became clear from going through several print-and-test cycles how crucial it is to validate designs in CAD software before committing to production in order to save time and material.

Arjun Rajesh

Throughout this project, I gained a much better understanding of how designs that work theoretically do not always behave the same way in real life. Although our calculations suggested that the robot would perform reliably, testing showed differences in actual performance, highlighting the need to validate designs through practical implementation. This made me realise the importance of debugging and adapting the system, rather than relying purely on theoretical results. I particularly enjoyed the CAD modelling sessions, where we worked on refining the arm design and ensuring the dimensions and workspace were suitable for the task. I also developed an appreciation for the 3D printing process, understanding how design decisions impact the ability to manufacture and assembly. This experience also helped me improve my coding skills, and my understanding of how different subsystems integrate to make a functioning robot.